

Mohamed Harmanani

Ph.D. Student, Vector Institute/Queen's University
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EDUCATION

Queen's University & Vector Institute

PhD. in Computer Science (Focus in AI)

Kingston, Canada

2024 – Present

- **Supervisor:** Prof. Parvin Mousavi

Queen's University & Vector Institute

MSc. in Computer Science (Focus in AI)

Kingston, Canada

2022 – 2024

- **Supervisor:** Prof. Parvin Mousavi
- **Thesis:** Towards a Reliable Deep Learning Framework for Prostate Cancer Diagnosis using Ultrasound

University of Toronto

BSc. (Hons.) in Computer Science

Toronto, Canada

2016 – 2021

- **Minor:** Philosophy

EXPERIENCE

Vector Institute

Doctoral Researcher

Toronto, Canada

2024 – Present

- **Supervisor:** Prof. Parvin Mousavi
- Designing and implementing trustworthy deep learning methods for cancer diagnosis from medical images using **vision-language and reasoning models**

Vector Institute

Graduate Research Assistant

Toronto, Canada

2022 – 2024

- **Supervisor:** Prof. Parvin Mousavi
- Designed an integrated system for clinically applicable detection of prostate cancer using **uncertainty estimation**
- Improved detection of prostate cancer from weakly labelled ultrasound data by 15% using **self-supervised learning**
- Developed a **novel multi-objective learning algorithm** for ultrasound-based cancer detection, improving performance by 10% over the baseline

Flinks

Data Scientist

Montréal, Canada

Sep 2021 – Sep 2022

- Accelerated the development of MLOps tools using DVC & Google Dataflow ETL to build data processing pipelines for NLP
- Optimized deep learning models (BERT) for financial language modelling using PyTorch and AWS SageMaker, resulting in a 33% increase in performance metrics

University of Toronto

Research Intern

Toronto, Canada

May 2021 – Sep 2021

- **Supervisor:** Prof. Lisa Zhang
- Implemented **deep learning** methods (RNN, BERT) for **natural language processing** (NLP) and **automated bug detection** in student code

University of Toronto

Research Assistant

Mississauga, Canada

Sep 2020 – May 2021

- **Supervisor:** Prof. Katharina Bräutigam
- Implemented a data pipeline for transcriptome analysis using Slurm, Bash and Python, reducing task turnaround time from 1 week to 2 days

Vennge

Software Engineer

Toronto, Canada

May 2019 – May 2020

- Implemented probabilistic algorithms for automated design generation in TypeScript, React & Fabric.js

1. **M. Harmanani**, B. Long★, Z. Guo★, P.F.R. Wilson, A. Sabour, M.N.N. To, G. Fichtinger, P. Abolmaesumi†, P. Mousavi†.
Vision-Language Models Encode Clinical Guidelines for Concept-Based Reasoning in Medical Image Analysis.
Under Review: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
2. M.N.N. To, D. Kim★, **M. Harmanani**★, P.F.R. Wilson, F. Fooladgar, S. Sojoudi, A. Jamzad, S. Abdalla, T. Tsang, C. Luong, S. Chang, P. Black, M. Leveridge, R. Siemens, R.G. Krishnan, P. Mousavi†, P. Abolmaesumi†.
Shift Happens: A Fairness-Oriented Framework for Medical Classification under Hidden Bias.
Accepted: Information Processing in Computer-Assisted Interventions conference (IPCAI), 2026.
3. P.F.R. Wilson, **M. Harmanani**, M.N.N. To, P.F.R. Wilson, A. Jamzad, T. Elghareb, Z. Guo, A. Kinnaird, B. Wodlinger, P. Abolmaesumi†, P. Mousavi†.
ProstNFound+: A Prospective Study using Medical Foundation Models for Prostate Cancer Detection.
Accepted: Int. J. Computer Assisted Radiology and Surgery (IJCARS), 2026.
4. Z. Guo, **M. Harmanani**, P.F.R. Wilson, M.N.N. To, T. Elghareb, O. Dzikunu, N. Maghsoodi, P. Abolmaesumi†, P. Mousavi†.
Pathology-Guided Contrastive Pretraining Enriches Preoperative CT Representations for Prognosis.
Accepted: SPIE Medical Imaging conference, 2026.
5. N. Radhakrishnan, S. Whitehall, **M. Harmanani**, B. Long, B. Zevin, R. Hisey.
Conductive-Thread Sensing for Open Inguinal Hernia Repair: A Low-Cost Interactive Phantom.
Accepted: SPIE Medical Imaging conference, 2026.
6. T. Elghareb, **M. Harmanani**★, M.N.N. To★, P.F.R. Wilson, A. Jamzad, F. Fooladgar, B. Abdelsamad, O. Dzikunu, S. Sojoudi, G. Reznik, M. Leveridge, R. Siemens, S. Chang, P. Black, P. Mousavi†, P. Abolmaesumi†.
ProTeUS: A Spatio-Temporal Enhanced Ultrasound-Based Framework for Prostate Cancer Detection.
Medical Image Computing and Computer Assisted Intervention conference (MICCAI), 2025.
7. **M. Harmanani**, A. Jamzad★, M.N.N. To★, P.F.R. Wilson★, Z. Guo, F. Fooladgar, S. Sojoudi, M. Gilany, S. Chang, P. Black, M. Leveridge, R. Siemens, P. Abolmaesumi†, P. Mousavi†.
Cinepro: Robust Training of Foundation Models for Cancer Detection in Prostate Ultrasound Cineloops.
IEEE International Symposium on Biomedical Imaging (ISBI), 2025.
8. **M. Harmanani**, P.F.R. Wilson, M.N.N. To, M. Gilany, A. Jamzad, F. Fooladgar, B. Wodlinger, P. Abolmaesumi★, P. Mousavi★.
TRUSWorthy: Towards Clinically Applicable Deep Learning for Confident Detection of Prostate Cancer in Micro-Ultrasound.
Int. J. Computer Assisted Radiology and Surgery (IJCARS), 2025.
9. M.N.N. To, P.F.R. Wilson, V. Nguyen, **M. Harmanani**, M. Cooper, F. Fooladgar, P. Abolmaesumi, P. Mousavi, R. Krishnan.
Improving Robustness to Subpopulation Shifts by Heuristic Subspace Exploration with Enhanced Diversification.
International Conference on Machine Learning (ICML), 2025.
10. P.F.R. Wilson, M.N.N. To, A. Jamzad, M. Gilany, **M. Harmanani**, T. Elghareb, F. Fooladgar, B. Wodlinger, P. Abolmaesumi, P. Mousavi.

ProstNFound: Integrating Foundation Models with Ultrasound Domain Knowledge and Clinical Context for Robust Prostate Cancer Detection.

Medical Image Computing and Computer Assisted Intervention conference (MICCAI), 2024.
(Early Accept, Top 11%)

11. P.F.R. Wilson, **M. Harmanani**, M.N.N. To, M. Gilany, A. Jamzad, F. Fooladgar, B. Wodlinger, P. Abolmaesumi, P. Mousavi.
Toward Confident Prostate Cancer Detection using Ultrasound: A Multi-Center Study.
Int. J. Computer Assisted Radiology and Surgery (IJCARS), 2024.
(Best Paper Runner-up)
12. M.N.N. To, F. Fooladgar*, P.F.R. Wilson*, **M. Harmanani***, M. Gilany, A. Jamzad, S. Sojoudi, S. Chang, P. Black, P. Mousavi[†], P. Abolmaesumi[†].
LensePro: Label Noise-Tolerant Prototype-Based Network for Improving Cancer Detection in Prostate Ultrasound with Limited Annotations.
Information Processing in Computer-Assisted Interventions conference (IPCAI), 2024.
13. M. Gilany, **M. Harmanani**, P.F.R. Wilson, A. Jamzad, M.N.N. To, B. Wodlinger, P. Abolmaesumi, P. Mousavi.
Calibrated Diverse Ensemble Entropy Minimization for Robust Test-Time Adaptation in Prostate Cancer Detection.
International Workshop on Machine Learning in Medical Imaging (MLMI) — Medical Image Computing and Computer Assisted Intervention conference (MICCAI), 2024.
14. **M. Harmanani**, P.F.R. Wilson, F. Fooladgar, A. Jamzad, M. Gilany, M.N.N. To, B. Wodlinger, P. Abolmaesumi, P. Mousavi.
Benchmarking Image Transformers for Prostate Cancer Detection from Ultrasound Data.
SPIE Medical Imaging conference, 2024.
15. **M. Harmanani.**
Modelling the Spread of COVID-19 in Indoor Spaces using Probabilistic Automated Planning.
Scheduling and Planning Applications woRKshop (SPARK) — International Conference on Automated Planning and Scheduling (ICAPS), 2023.
16. S. Fujimori, **M. Harmanani**, O. Siddiqui, L. Zhang.
Using Deep Learning to Localize Errors in Student Code Submissions.
ACM Technical Symposium on Computer Science Education (SIGCSE), 2022.

TEACHING

- **Queen's University**, Teaching Assistant
- CISC 365 - Algorithms I Winter 2026, Winter 2024
- CISC 881 - Medical Imaging and Machine Learning Fall 2025
- BMIF 801 - Programming for Biomedical Computing Spring 2025
- CISC 325 - Human-Computer Interaction Winter 2025
- CISC 452 - Neural and Genetic Computing Fall 2024, Fall 2023
- CISC 151 - Intro to Data Analytics Winter 2023

AWARDS

Academic Honours and Scholarships:

- **Vector Institute Research Grant II** 2025
- **Bruce Mitchell Research Fellowship**, Queen's University 2024-Present
Awarded to doctoral trainees "above and beyond the normal graduate trainee complement"
- **Vector Institute Research Grant I** 2023, 2024

- **Robert Sutherland Fellowship**, Queen's University 2022-2023
Awarded to outstanding students belonging to a visible minority
- **Undergraduate Research Award**, University of Toronto 2022
- **Undergraduate Entrance Award**, University of Toronto 2016

Media:

- **BetaKit**: Making AI work beyond the lab May 2025

ACADEMIC SERVICE

Reviewer:

- CVPR: IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026
- ImNO: Imaging Network of Ontario Symposium 2026
- ISBI: IEEE International Symposium on Biomedical Imaging 2026, 2025
- IPCAI: Information Processing in Computer-Assisted Interventions 2026
- IJCARS: International Journal of Computer Assisted Radiology and Surgery 2025
- MICCAI: Medical Image Computing and Computer-Assisted Interventions 2025
- IGTxImNO Joint Symposium on Medical Imaging, Image-Guided Therapies, and AI 2025

Leadership:

- **Seminar Coordinator**, Med-i Lab, Queen's University Sep 2023 - Dec 2025
Organizing the lab's weekly meetings and seminars, and facilitated discussion of key topics

SKILLS

Programming: Python, PyTorch, C, SQL, R, MATLAB, Java, JavaScript, HTML, CSS

Tools: PyTorch, Keras, BigQuery, Google Dataflow, GCP, AWS (S3, SageMaker), React, Node.js, Seaborn

Algorithms: CNN, ResNet, Transformers, Vision Transformers, Large Language Models (BERT, GPT)

LANGUAGES

English (fluent), French (fluent), Arabic (fluent), Spanish (intermediate)